Zeran Ni (Simon)

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Education

New York University, Center For Data Science, College of Arts and Science *Sep 2023 – May 2027*
Major: **Computer Science and Data Science**
GPA: 3.93/4.0
Relevant Courses: Data Structures and Algorithms, Fundamentals of Machine Learning, Software Engineering

Experience

Research Intern *New York, NY*
RiskEcon® Lab, Courant Institute of Mathematical Sciences, NYU *Dec 2025 – Present*

- Built county-to-weather-zone and wind/solar region mappings by reconciling mismatched identifiers across four public datasets.
- Analyzed relationships between hourly load and heating/cooling degree days (HDD/CDD) using 3D histograms, load duration curves, and seasonal heatmaps.

Class Tutor and Grader *New York, NY Jan 2024 – Dec 2025*
Department of Computer Science, New York University

- Led **3** hours of in-class tutoring and **7** hours of office hours each week, supporting **200+** students with Python programming (control flow, OOP, File I/O, and data structures) and assignments debugging.
- Graded **11** programming assignments for ∼**180** students, delivering detailed and constructive feedback.

Projects

AI-Assisted Autograder for Introductory CS Course at NYU *GitHub Repository*

- Designed an AI-powered autograder evaluating 11 programming assignments for 180+ students, reducing grading time by over 60% by integrating OpenAI GPT models via LangChain to assess code quality, formatting standards, and programming style beyond traditional unit tests.
- Engineered dynamic testing pipeline using `subprocess` for code execution, `re` for output validation, and custom parsers to extract numerical values from formatted tables, with adaptive input handling that automatically feeds test cases and flags edge cases requiring human review.
- Overcame Gradescope's function-call dependency limitations and assess early assignments before students learned advanced programming concepts.
- Technologies Used: `Python, OpenAI API, LangChain, Bash, Regular Expression, Git, VS Code.`

Pymodoro *GitHub Repository*

- Architected a daemon Pomodoro timer that runs invisibly while the user keeps using their terminal.
- Designed a file-based IPC layer so the user-facing CLI (`--show`, `--status`, `--stop`) communicates with the background daemon without blocking the terminal.
- Published to PyPI via GitHub Actions CI on Python 3.12/3.13 (Linux and Windows).
- Technologies Used: `Python, pytest, PyPI, CLI, GitHub Actions.`

Tetruzzle *GitHub Repository*

- Built a Dockerized computer vision app that reads real Tetris board states from screenshots using OpenCV and NumPy, with Hough line detection for board boundary extraction and color-based mino classification across 10×20 cells.
- Developed REST APIs to handle requests between databases, containerized Flask and ML services, with a CI/CD pipeline via GitHub Actions that builds, tests, and deploys to DigitalOcean on every push.
- Technologies Used: `Python, OpenCV, NumPy, Flask, Docker, GitHub Actions, DigitalOcean.`

Technologies

Core Languages: Python (Primary), Java (Proficient), JavaScript, C, SQL, HTML5/CSS3
Libraries: NumPy, Pandas, PyTorch, Scikit-learn, SciPy, LangChain, Flask
Tools & Platforms: Git, Github, Github Actions, Docker, MongoDB, Jupyter Notebook, VS Code, Claude code, IntelliJ IDEA, PyPI, Cloudinary, Vercel, DigitalOcean, Overleaf